Answer **one** question from this section in the spaces provided.

**8** **(a)** A small steel sphere is oscillating vertically on a spring, as shown in Fig. 8.1.

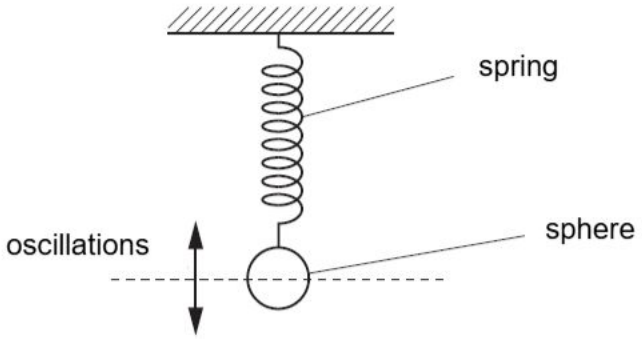



**Fig. 8.1**

Fig. 8.2 shows the variation with displacement $x$ of the energy $E$ of the oscillations.

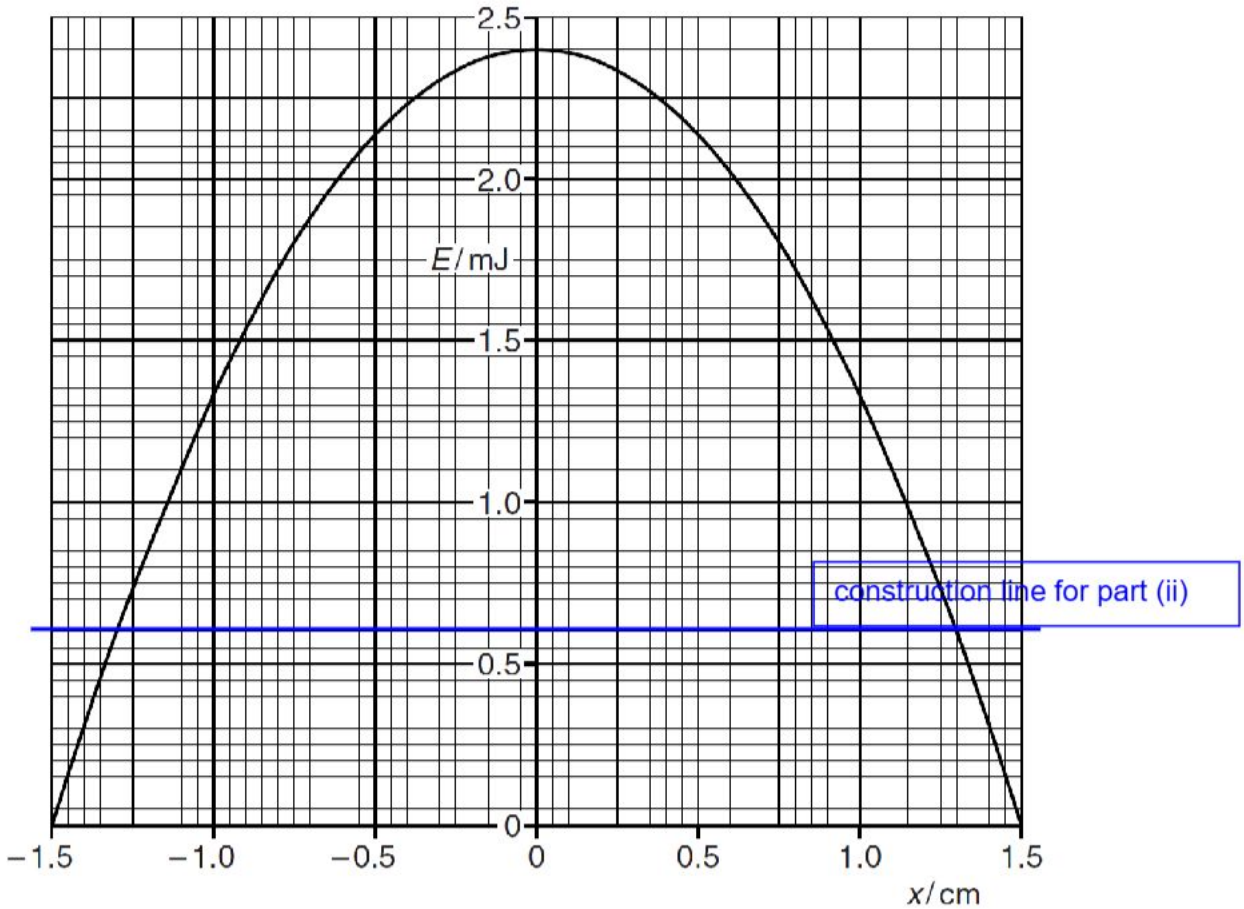



**Fig. 8.2**

**(i)** State whether the energy $E$ represents the total energy, the potential energy or the kinetic energy of the oscillations.

Kinetic energy [B1] .....[1]

**(ii)** The amplitude of the oscillations reduces over a long period of time. After many oscillations, the maximum value of $E$ decreases **by** 0.60 mJ.

Without further calculation, use Fig. 8.2 to determine the new amplitude of the oscillations. Show your construction.

suitable line that shifts the $x$-axis up by 0.60 mJ to intersect the original curve [B1]
Also accept suitable parabolic sketch that shifts down the peak and all points on original curve by 0.60 mJ to intersect the $x$-axis

→ new amplitude (intersection) of **1.3 cm** [B1]

new amplitude = ..... cm [2]

**(b)** The steel sphere is held between two fixed points A and B by two identical springs, as shown in Fig. 8.3.

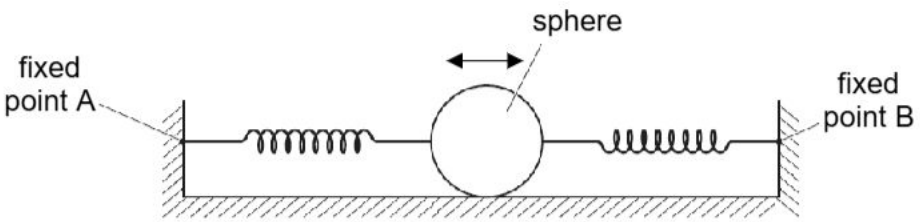



**Fig. 8.3**

The sphere is displaced from its equilibrium position and then released so that it is free to oscillate horizontally along the line AB. During the oscillations, the springs remain extended and do not exceed their limits of proportionality.

The resultant force acting on the sphere is given by

$$F = -2kx$$

where $k$ is the spring constant and $x$ is the displacement of the sphere from equilibrium position.

**(i)** Explain how the equation shows that the oscillations of the sphere are simple harmonic.

- $k$ is a (positive) constant so $F$ is (directly) proportional to $x$
- Negative sign means $F$ is directed opposite of $x$

Since $F$ is (directly) proportional and in the same direction as acceleration $a$, then $a$ is proportional and directed opposite to $x$, hence simple harmonic motion.

[B1] either one of two conditions stated [2]

[B1] both conditions stated AND link back to $F \propto a \propto -x$

**(ii)** Show that the frequency $f$ of the oscillations is given by

$$f = \frac{1}{2\pi}\sqrt{\frac{2k}{m}}$$

where $m$ is the mass of the sphere.

$F = ma$

$-2kx = ma$

$a = -\left(\frac{2k}{m}\right)x$ [M1]

$\omega^2 = \frac{2k}{m}$ leading to $\omega = \sqrt{\frac{2k}{m}}$ and $\omega = 2\pi f$ [A1]

so $f = \frac{1}{2\pi}\sqrt{\frac{2k}{m}}$

[2]

The variation of the acceleration $a$ of the sphere with its displacement $x$ from the equilibrium position is shown in Fig. 8.4.

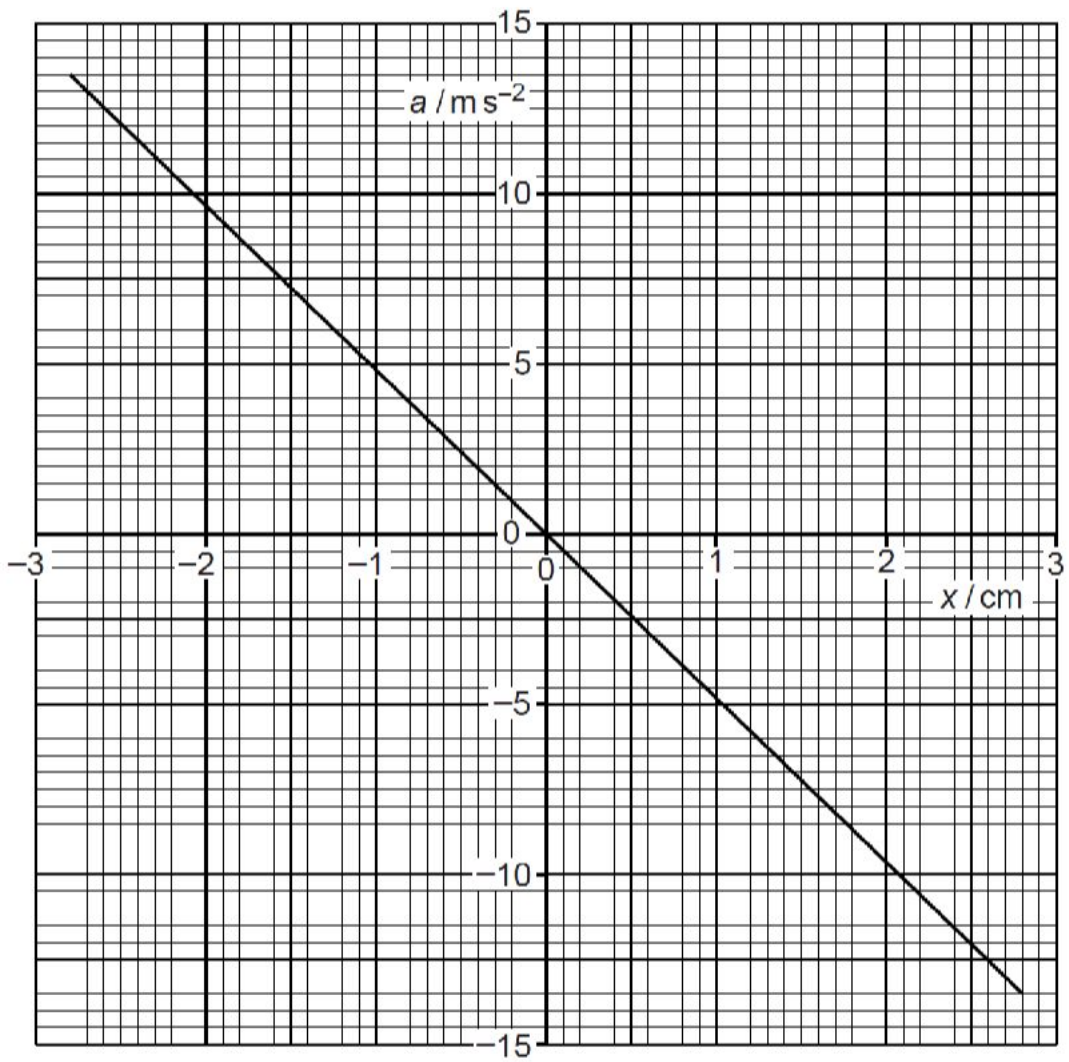



**Fig. 8.4**

**(iii)** Use Fig. 8.4. to determine the frequency of oscillations of the sphere.

gradient $= -\omega^2$ [C1]

$$\omega^2 = -\frac{-13.5 - 13.5}{2.8\times10^{-2} - \left(-2.8\times10^{-2}\right)} = 482$$

$\omega = 22.0$ rad s$^{-1}$ [C1]

$$f = \frac{22.0}{2\pi} = 3.5 \text{ Hz}$$ [A1]

$f =$ ..... Hz [3]

**(iv)** State **one** way how the period of oscillations of the sphere can be reduced.

Use a sphere of lighter mass (reduce $m$) OR use a stiffer spring (increase $k$) [B1]

NOT damping or increasing the amplitude [1]

**(v)** The variation of the displacement of the sphere with time is sinusoidal. The variation of the velocity of the sphere with time is also sinusoidal.

**1.** State the phase difference between the displacement and the velocity of the sphere.

phase difference = 90° [B1] ..... ° [1]

**2.** The sphere is released from its maximum positive displacement at time $t = 0$.

Calculate the magnitude of the velocity of the sphere at time $t = \frac{T}{12}$, where $T$ is the period of oscillations.

$x = x_0 \cos \omega t$

$$x = \left(2.8\times10^{-2}\right)\cos\left(\frac{2\pi}{T}\times\frac{T}{12}\right)$$ [M1]

$x = 0.0242$ m

$v = \omega\sqrt{x_0^2 - x^2}$

$$= 22\sqrt{\left(2.8\times10^{-2}\right)^2 - 0.0242^2}$$ [M1]

$= 0.31$ m s$^{-1}$ [A1]

velocity = ..... m s$^{-1}$ [3]

**(vi)** The microscopic structure of a solid crystal can be pictured as planes of atoms, as shown in Fig. 8.5. The distance between the centres of neighbouring atoms is indicated.

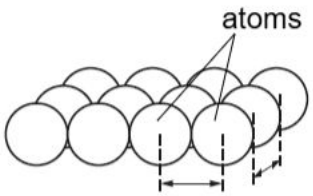



**Fig. 8.5**

Suggest why the spring-sphere system in Fig. 8.3 provides a simple model for the motion of an atom in a solid crystal.

Sphere can be modelled as an atom oscillating between adjacent atoms in the crystal, where the (restoring) spring forces resemble the attractive and repulsive interatomic forces when the atom is displaced slightly further apart or closer to adjacent atoms, respectively [B1] [1]

**(c)** The fixed point B is now replaced with an oscillator that causes horizontal oscillations of the sphere, as shown in Fig. 8.6. Both springs remain extended within the limit of proportionality.

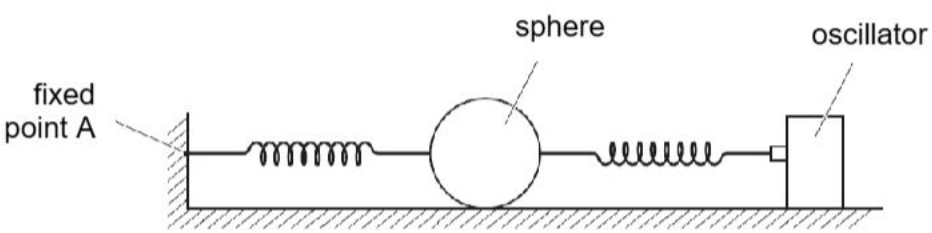



**Fig. 8.6**

The oscillator is switched on. The amplitude of vibration of the oscillator is constant. The frequency $f$ of vibration of the oscillator is varied.

The variation of the amplitude $x_0$ of the oscillations with frequency $f$ of the oscillator is shown in Fig. 8.7.

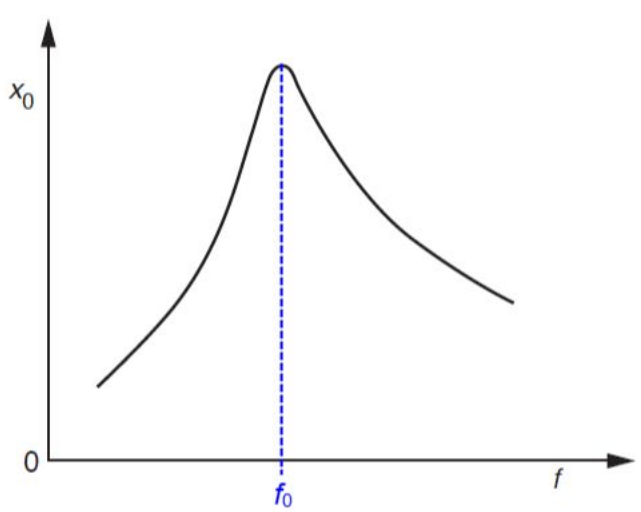



**Fig. 8.7**

**(i)** Explain why there is a peak in Fig. 8.7 and state the frequency at which the peak occurs.

The peak suggests that resonance occurred since the sphere is under forced oscillations where the oscillator's driving frequency is equal to the natural frequency of the system [B1]

The frequency at which the peak occurs = 3.5 Hz OR ecf their value of *f* [B1]

[2]

**(ii)** State and explain **one** modification to the setup in Fig. 8.6 to reduce the amplitude of the oscillations in Fig. 8.7.

- EITHER idea of increased damping by a sensible method (e.g. attaching a light card with large surface area to increase air resistance, sprinkling sand on the surface to increase friction) [B2]
- OR reduce amplitude of vibration of oscillator to reduce energy input to system (B2) [2]

[Total: 20]